Autoencoders

FRC-EVL

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Encoder–Decoder Architecture: A Powerful Paradigm

The encoder–decoder structure is one of the most powerful ideas in Deep Learning.

Core Idea

Encode information into a meaningful latent representation and then decode it into a desired output.

- Used in **Autoencoders** for representation learning.
- Used in **Machine Translation** and sequence-to-sequence models.
- Foundation of the **Transformer architecture**.
- Used in **Large Language Models (LLMs)**.

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Why is it so powerful?

It separates:

- Understanding / Representation learning (Encoder)
- Generation / Reconstruction (Decoder)

This modularity enables flexible, scalable, and generative systems.

Autoencoders

- Autoencoders are a special type of fully-connected neural networks.
- Try to reproduce the input data at the output of the network.
- Architecture that allows very interesting tasks.

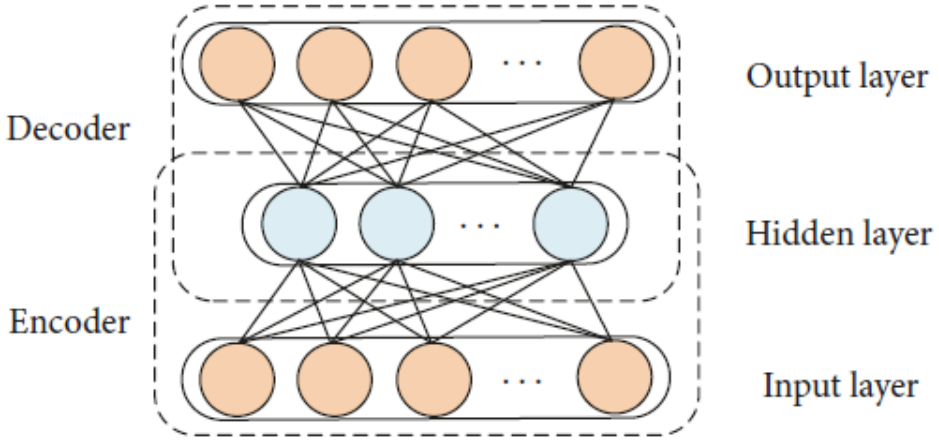
Three main applications:

- **Dimensionality reduction:** Learn an efficient reduced representation of unlabeled data.
- **Neural network pretraining:** Detect relevant attributes and pretrain networks.
- **Synthetic data generation:** Generate data similar to input data to enlarge training datasets.

Basic Structure

An autoencoder has two main parts:

- **Encoder:** Converts inputs into a latent representation (latent space), usually lower dimensional.
- **Decoder:** Reconstructs the input from the latent representation.



Pre-training: Learning Rich Deep Features

One of the key advantages of autoencoder pre-training is the **automatic extraction of high-level features**.

Deep Feature Learning

The encoder does not simply reduce dimensionality — it learns structured, informative representations of the data.

- Captures nonlinear relationships between variables.
- Extracts hierarchical representations.
- Filters noise and redundancy.
- Produces compact but expressive latent features.

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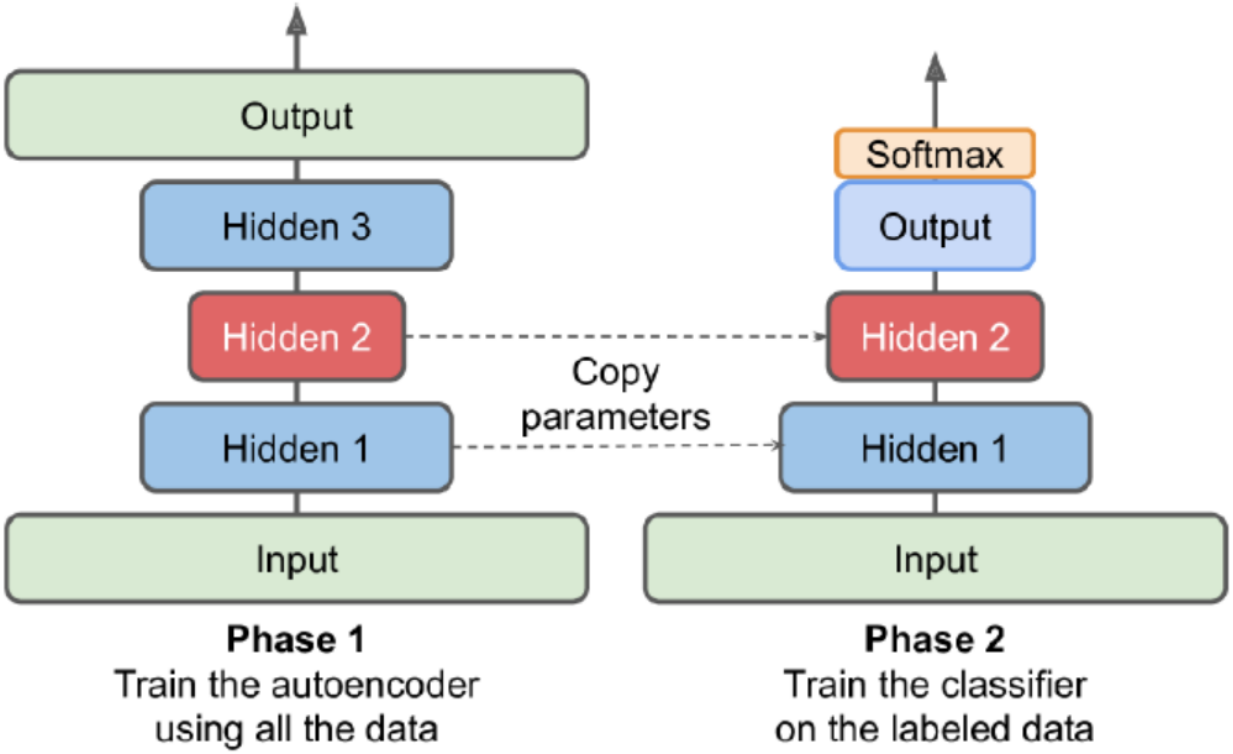
Impact on the Classifier

The classifier no longer works on raw input features, but on **deep, semantically meaningful representations**, which often improves generalization.

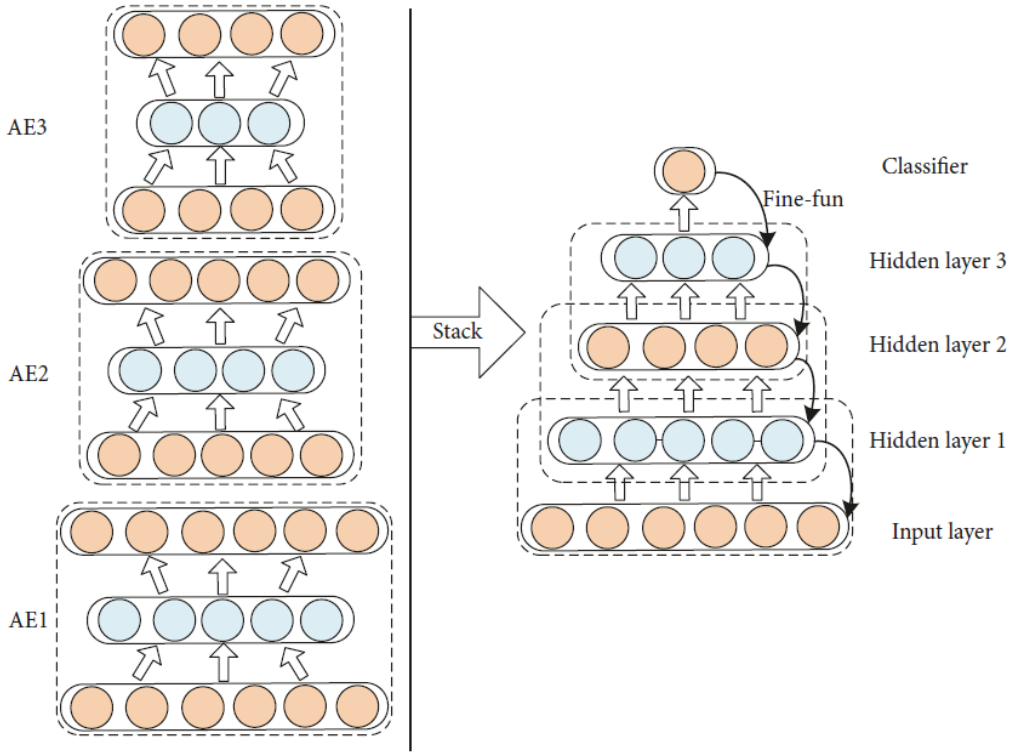
Pre-training Procedure

1. Train the autoencoder using all available data (labeled or unlabeled).
2. Copy weights and biases up to the latent layer.
3. Build a classifier on top of the latent layer.
4. Train the classifier using labeled data.
5. Usually, pretrained weights are kept frozen.



Stacked Autoencoders (SAE)

SAEs stack n autoencoders into n hidden layers using unsupervised layerwise training, followed by supervised fine-tuning.



1. Train the first autoencoder and obtain learned features.
2. Use learned features as input for the next autoencoder.
3. Repeat until all hidden layers are trained.
4. Apply backpropagation with labeled data for fine-tuning.

Why SAEs Shine with Few Labeled Data

Key Advantage

They decouple representation learning from supervised learning.

When labeled data is scarce:

- Train the SAE using all available (unlabeled) data.
- Learn rich, hierarchical representations.
- Train the classifier using only a small labeled subset.

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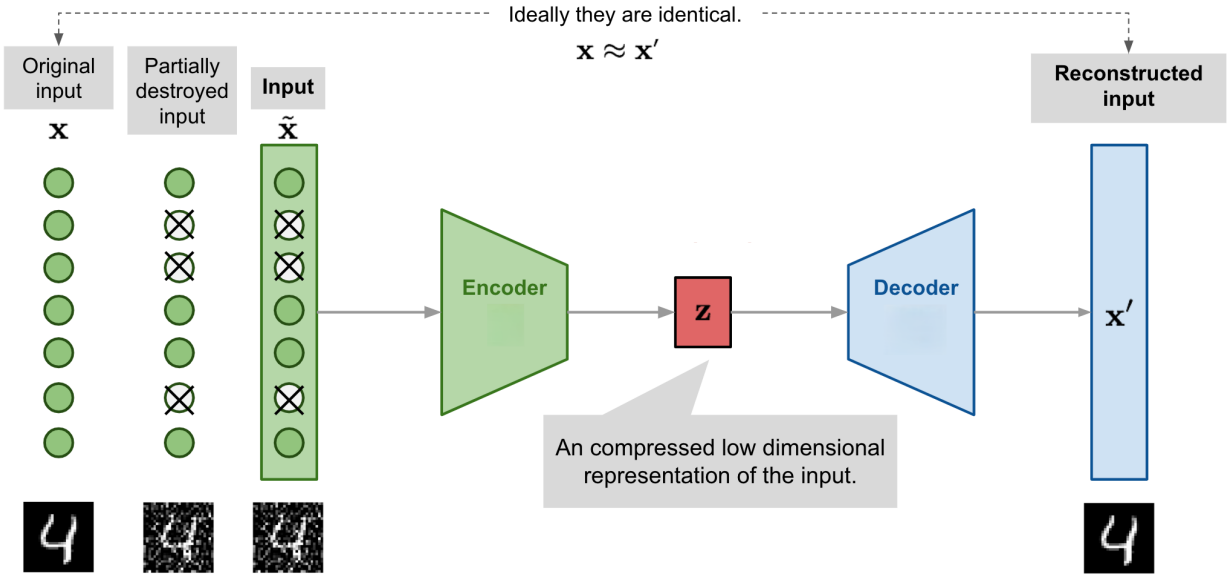
Benefits:

- Reduces dimensionality.
- Mitigates overfitting.
- Improves generalization.

Denoising Autoencoder

A modification designed to prevent learning the identity function.

- Corrupt input data by adding noise or masking inputs.
- Compare output with original (non-corrupted) input.
- Forces extraction of robust features.



Variational Autoencoders (VAE)

Standard autoencoders:

- Learn compact representations.
- Latent space may not be continuous.

VAEs (We will see them in Unit 4):

- Latent space is continuous by design.
- Allow random sampling.
- Enable smooth interpolation.
- Useful for generative modeling.

Summary

- Autoencoders learn compressed representations.
- Useful for dimensionality reduction and pretraining.
- Variants include:
  - Stacked Autoencoders
  - Denoising Autoencoders
  - Variational Autoencoders